Axiomatic Theory of Design

May 2026

0.1 Property

Property is an ordered pair:

$$\vec{x} = (x_n, x_v)$$

where $x_n \in D_x$ and $x_v \in R_x$.

0.2 Property set

Property set contains every possible property:

$$X \subseteq D_x \times R_x$$

$$\vec{x} \in X$$

0.3 Object

Object is a set of properties:

$$O = \{x_1, x_2, \ldots\} \subseteq X$$

possibly infinite, and even uncountable?

0.4 Structure

Structure:

$$\oplus O = O \cup (O \times O)$$

Some laws:

$$\oplus O = \oplus(O_1 \cup \cdots \cup O_n) = \bigcup_i (\oplus O_i) \cup \bigcup_{i \neq j} (O_i \times O_j)$$

0.5 Universe

Universe-environment-product:

$$U = E \cup S$$

Some laws:

$$\oplus U = (\oplus E) \cup (\oplus S) \cup (E \times S) \cup (S \times E)$$

0.6 Performance

Performance:

$$P \subseteq A \times R$$

where $A \subseteq E \times S$ and $R \subseteq S \times E$.

0.7 Design requirement

$$r_d : S \rightarrow [S]$$

where $[S]$ is the goal/constraint.

0.8 Design evaluation

$$\oplus(E \cup \bar{S} \cup S') = \oplus(E' \cup S')$$